

Programming In C-HDOC DAY_17

Problem

Write a menu-driven C program using **switch-case** to perform digit-based operations on a positive integer. Define **one separate user-defined function for each menu option**, *with void return type and with a simple function call (without call by value/reference)*. Each function should read the required input, process it, and display the result. Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

- Generate the largest possible number using its digits
- Generate the smallest possible number using its digits.
- Swap its first and last digits.
- Check whether it is a palindrome number.

ALGORITHM

Step 1: Start

1. Start the program.
2. Declare a character array num1[100] and an integer variable c.
3. Ask the user to enter a positive integer and store it as a string in num1.
4. Display the menu containing four choices:
 - Generate the largest number using the digits.
 - Generate the smallest number using the digits.
 - Swap the first and last digits.
 - Check whether the number is a palindrome.
5. Read the user's choice c.

Step 2: Perform the selected operation

Use a switch statement based on c.

Case 1: Find largest number

1. Find the length of the string.
2. Use bubble sort to arrange the digits in **descending order**.
3. Compare adjacent characters.
4. If the left character is smaller than the right character, swap them.
5. Repeat until all digits are sorted.

6. Display the resulting string as the largest number.

Example:

Input: 35281

Output: 85321

Case 2: Find smallest number

1. Find the length of the string.
2. Use bubble sort to arrange the digits in **ascending order**.
3. Compare adjacent characters.
4. If the left character is greater than the right character, swap them.
5. Repeat until all digits are sorted.
6. Display the resulting string as the smallest number.

Example:

Input: 35281

Output: 12358

Case 3: Swap first and last digit

1. Find the last index of the string.
2. Set start = 0.
3. Set end = strlen(n) - 1.
4. Swap the characters at start and end.
5. Display the resulting number.

Example:

Input: 12345

Output: 52341

Case 4: Check palindrome

1. Find the length of the input string.
2. Create another character array n1.
3. Copy the characters of n into n1 in reverse order.
4. Add '\0' at the end of n1.
5. Compare n and n1 using strcmp().
6. If they are equal, display "**number is palindrome**".
7. Otherwise, display "**number is not palindrome**".

Example:

Input: 12321

Reverse: 12321

Output: Number is palindrome

Step 3: Invalid choice

If c is not between 1 and 4, display:

Invalid choice.

Step 4: Stop

C PROGRAM CODE

```
#include <stdio.h>
#include <string.h>

void largest(char n[100])
{
    int len=strlen(n);
    for(int i=0;i<(len-1);i++)
    {
        for(int j=0;j<(len-i-1);j++)
        {
            if(n[j]<n[j+1])
            {
                char temp=n[j];
                n[j]=n[j+1];
                n[j+1]=temp;
            }
        }
    }
    printf("the largest number which can be made by digits of entered number is %s",n);
}

void smallest(char n[100])
{
    int len=strlen(n);
    for(int i=0;i<(len-1);i++)
    {
        for(int j=0;j<(len-i-1);j++)
        {
            if(n[j]>n[j+1])
            {
```

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```
        char temp=n[j];
        n[j]=n[j+1];
        n[j+1]=temp;
    }
}

printf("the smallest number which can be made by digits of entered number is %s",n);
}

void swap(char n[100])
{
    int end=strlen(n)-1;
    int start=0;
    while(n[start]!=' ')
        start++;
    while(n[end]!=' ')
        end--;
    char temp=n[start];
    n[start]=n[end];
    n[end]=temp;
    printf("the number is %s",n);
}

void palindrome(char n[100])
{
    int len=strlen(n);
    char n1[len+1];
    for(int i=len-1, j=0; i>=0; i--, j++)
    {
        n1[j]=n[i];
    }
    n1[len]='\0';
    if(strcmp(n,n1) == 0)
        printf("number is palindrome");
    else
        printf("number is not palindrome");
}
```

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```
int main()
{
    char num1[100];int c;
    printf("enter a positive integer\n");
    scanf("%s", num1);
    printf("choice 1 is to generate the largest number using digits of positive integer entered\n");
    printf("choice 2 is to generate the largest number using digits of positive integer entered\n");
    printf("choice 3 is to swap the last and first digit off number entered\n");
    printf("choice 4 is to check wether the entered num is palindrome\n");
    printf("enter choice\n");
    scanf("%d", &c);
    switch(c)
    {
        case 1:
        {
            largest(num1);
            break;
        }
        case 2:
        {
            smallest(num1);
            break;
        }
        case 3:{
            swap(num1);
            break;
        }
        case 4:
        {
            palindrome(num1);

            break;
        }
        default:
            printf("Invalid choice.\n");
    }
```

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```
    break;  
}  
return 0;  
}
```